

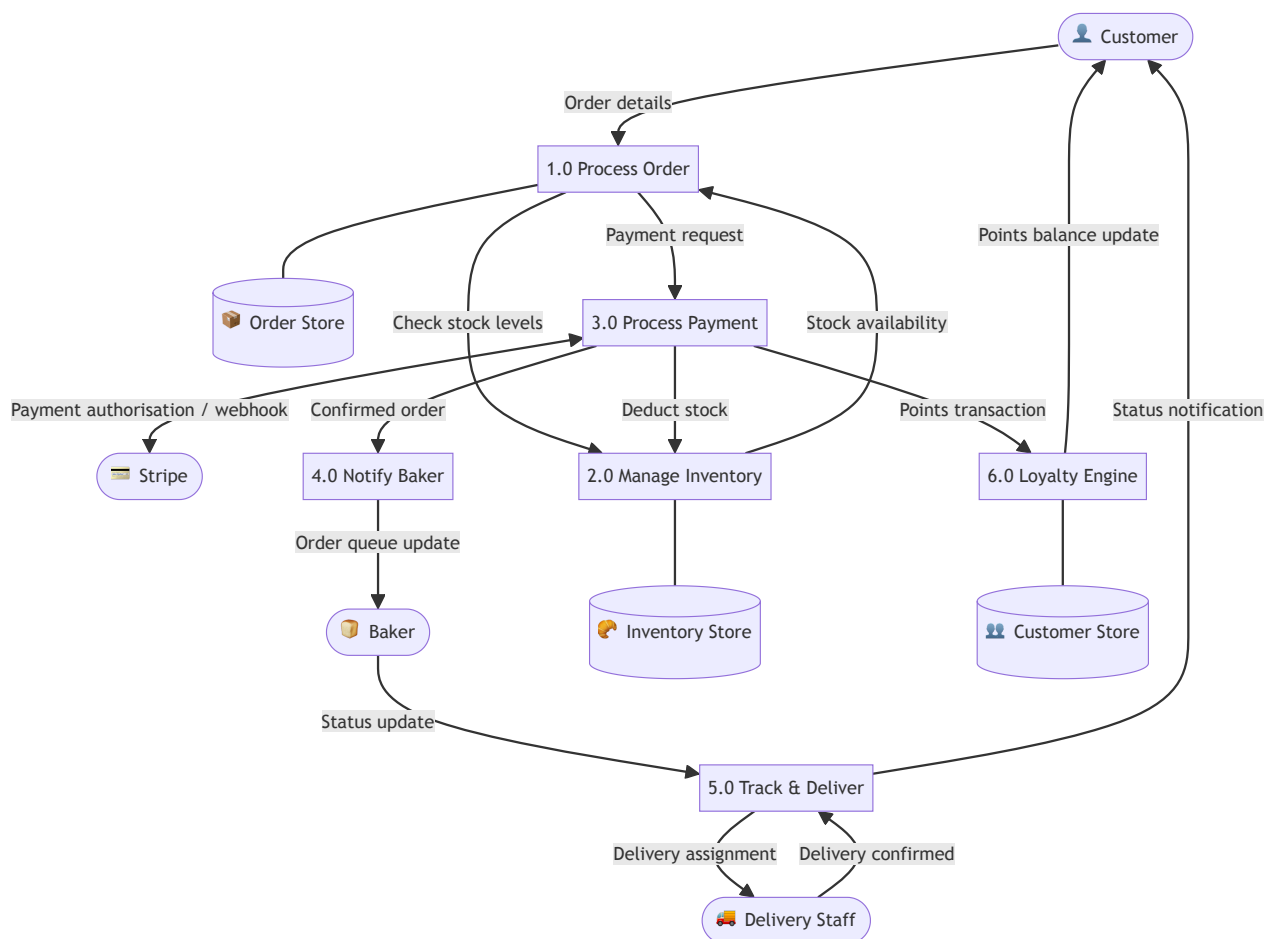
Data Flow Diagram — Level 1 (Order Flow Decomposition)

FreshBakes Bakery | IS501 Project

Level 1 decomposes the central system into its **six core processes**, showing how data flows between them, external entities, and data stores. This diagram traces the complete path of an order from customer submission through to delivery.

Processes: 1.0 Process Order · 2.0 Manage Inventory · 3.0 Process Payment · 4.0 Notify Baker · 5.0 Track & Deliver · 6.0 Loyalty Engine

Data Stores: Order Store · Inventory Store · Customer Store



Process Descriptions

Process	Responsibility	Key Inputs	Key Outputs
1.0 Process Order	Validates order details and checks stock before proceeding to payment	Order details from Customer	Payment request; stock check to Process 2.0
2.0 Manage Inventory	Tracks current stock; deducts on payment; raises low-stock alerts	Stock check requests; deduction triggers from Process 3.0	Stock availability; updated Inventory Store
3.0 Process Payment	Handles Stripe checkout flow and processes webhook confirmation	Payment request; payment_intent.succeeded from Stripe	Confirmed order to Baker; stock deduction trigger; loyalty transaction
4.0 Notify Baker	Pushes confirmed orders to the baker queue	Confirmed order from Process 3.0	Order queue update for Baker
5.0 Track & Deliver	Manages order status transitions and delivery assignment	Status updates from Baker; delivery confirmations from Staff	Delivery assignment to Staff; status notification to Customer
6.0 Loyalty Engine	Awards and redeems loyalty points per transaction	Points transaction from Process 3.0	Updated loyalty balance to Customer Store and Customer